

AutoResearch workflow

Literature retrieval
source spans

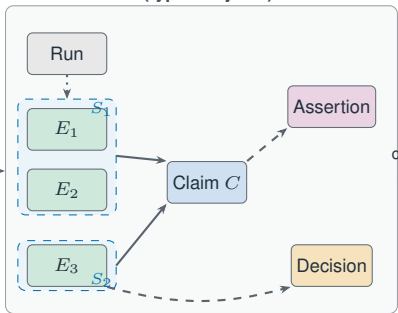
Experiments and runs
code, logs, artifacts

Analysis and decisions
claims, rationale

Manuscript drafting
sentences, tables

declared
relations

Research state (typed objects)



→ sufficient set - → depends_on... → derived_from

Revision event

reject · supersede · invalidate run

Transaction manager

1 Apply operations to a private copy of the head

2 **Propagate**: lineage closure → claims → assertions → decisions

3 **Check** $I_1 - I_6, I_8$: references, reciprocity, derived states, legal transitions

4 **Freshness** I_7 : head equals base hash?

Commit
new head

Roll back
head kept

on commit

Tamper-evident store

Content-addressed
snapshot (SHA-256)

Hash-chained
transaction log

History audit
recompute, verify HEAD

Export gate

every assertion admissible